

Corrine Sadler

D598-Analytics Programming

January 9, 2026

## **Task 2: Coding**

### **A. Python Program**

The Python program developed for Task 2 performs all required data analysis using the pandas library. The program imports the D598 dataset from an Excel file located in the same directory and verifies successful loading by previewing the data. Duplicate rows are identified to ensure data integrity. Businesses are grouped by Business State, and descriptive statistics (mean, median, minimum, and maximum) are calculated for all numeric variables in a separate dataframe.

The program also filters businesses with negative debt-to-equity ratios for further evaluation of their financial risk. Additionally, a new dataframe is created to calculate the debt-to-income ratio, which is defined as total long-term debt divided by total revenue. To prevent division-by-zero errors and preserve data integrity, an explicit conditional check is implemented so that businesses with zero total revenue are assigned a NaN value for the debt-to-income ratio. The calculated debt-to-income ratio is then concatenated back to the original dataset, as required by the task instructions. All codes executes without errors and follows Python best practices.

GitLab Link: [https://gitlab.com/wgu-gitlab-environment/student-repos/csadle18/d598-analytics-programming/-/blob/27a29794339546357aeb4dac5e35febd1cd29f03/D598\\_Task2\\_CorrineSadler\\_Attemp3.ipynb](https://gitlab.com/wgu-gitlab-environment/student-repos/csadle18/d598-analytics-programming/-/blob/27a29794339546357aeb4dac5e35febd1cd29f03/D598_Task2_CorrineSadler_Attemp3.ipynb)

## **Sources**

The only sources used were the course materials provided by WGU.

## **B. Professional Communication**

This submission demonstrates professional communication through clear code structure, meaningful comments, and consistent formatting. Code comments are included to explain each significant processing step, and variable naming conventions are used to enhance readability and understanding.